

# Cheng Xu

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**Research Interests:** VLA Models, Robotic Manipulation, World Models, Multimodal Perception, and Tactile Sensing

## Education

### The University of Tokyo

*M.Eng. in Electrical Engineering and Information Systems*

*Research:* Visual perception and multimodal sensor fusion for autonomous train localization.

Tokyo, Japan

*Apr. 2026 – Present*

### University of Sussex

*B.Eng. (Hons) in Robotics and Electrical Engineering, First-Class Honours*

Dual-degree program average: **88/100** | Rank: **Top 5%**

Brighton, United Kingdom

*Sep. 2021 – Jun. 2025*

### Zhejiang Gongshang University

*B.Eng. in Electronic Information Engineering*

Dual-degree program average: **88/100** | Rank: **Top 5%**

Hangzhou, China

*Sep. 2021 – Jun. 2025*

## Selected Experience

### AGIBOT Japan

*Robotics Engineering Intern*

Tokyo, Japan

*Jun. 25 – Jul. 3, 2026*

- Studied Physical Intelligence's official  $\pi_0$  source code, focusing on its model architecture and action-generation process.
- Assisted with the inspection, disassembly, and maintenance of TX-G2 humanoid robots.
- Assessed deployment requirements and technical challenges for a robotic deformable-object loading task in a factory environment.

## Selected Projects

### Tactile Sensors for Robotic Manipulators

*Oct. 2024 – Mar. 2025*

- Designed and evaluated an eight-layer,  $16 \times 16$  flexible piezoresistive tactile sensor array using Velostat and Arduino Nano row-column acquisition.
- Developed the readout circuit and PCB in EasyEDA and an Arduino-Python pipeline for real-time pressure visualization and four-stage noise reduction.

### Vision- and Multi-Sensor-Fusion-Based Localization for Autonomous Trains

*Apr. 2026 – Present*

- Developing a localization and velocity-estimation system for autonomous trains in GNSS-denied environments such as tunnels and railway depots.
- Investigating track, switch, and landmark recognition using semantic segmentation, object detection, and optical flow, followed by fusion with IMU, wheel odometry, LiDAR, and track-topology maps.

### VLA Fundamentals

*May 2026 – Present*

- Analyzed Physical Intelligence's official  $\pi_0$  PyTorch implementation, focusing on multimodal preprocessing, flow matching, and action denoising; currently preparing to reproduce and fine-tune  $\pi_{0.5}$  for robotic manipulation tasks.

## Technical Skills

**Programming & ML:** Python, PyTorch, C, MATLAB

**Hardware & Tools:** Arduino, EasyEDA, Multisim, schematic capture, PCB layout

## Honors & Awards

- **Outstanding Student Award (Top 1.5%)**, Zhejiang Gongshang University, 2024 – 2025
- **First-Class Scholarship (Top 1.5%)**, University of Sussex, 2024 – 2025
- **Academic Scholarship**, University of Sussex, 2022 – 2023

## Languages

**Japanese:** JLPT N1 | **English:** TOEFL iBT 87